SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct
I Gurusurya.G 192311332_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: Gurusurya___________

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy.
   How can encryption and access control protect sensitive backup data?
   Explain the cost advantages compared to traditional storage systems.
   Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. Cloud Auto-Scaling and Load Balancing for Seasonal Sales

Auto-Scaling

Cloud auto-scaling automatically increases or decreases the number of servers based on traffic demand. During seasonal sales, new instances are added to handle high user traffic and removed when demand decreases.

Load Balancing

A load balancer distributes incoming requests evenly across multiple servers, preventing any single server from becoming overloaded.

Benefits

- Handles sudden traffic spikes efficiently.
- Reduces downtime by preventing server overload.
- Improves user experience with faster response times and uninterrupted service.

Role of Monitoring Tools

Monitoring tools continuously track CPU usage, memory, network traffic, and response time. When predefined thresholds are reached, they trigger auto-scaling and send alerts to administrators.

Real-World Example

During Amazon Prime Day or Flipkart Big Billion Days, millions of users visit simultaneously. Auto-scaling launches additional servers, while load balancing distributes customer requests evenly, ensuring smooth shopping without website crashes.

Conclusion

Auto-scaling, load balancing, and monitoring together provide high availability, better performance, and a seamless customer experience during peak traffic.

2. Secure Cloud Storage for Backup Data

Durability and Data Redundancy

Cloud storage stores multiple copies of backup data across different servers or regions. If one storage device fails, another copy is available, ensuring high durability and reliability.

Encryption and Access Control

- Encryption: Data is encrypted both at rest (AES-256) and in transit (TLS/SSL).
- Access Control: Role-Based Access Control (RBAC) and Multi-Factor Authentication (MFA) ensure only authorized users can access backup data.

Cost Advantages

- No need to purchase physical storage hardware.
- Pay only for the storage used.
- Reduced maintenance and infrastructure costs.
- Automatic scalability as storage needs grow.

Disaster Recovery Example

If a company's primary data center is damaged due to flooding, backup data stored in the cloud can be restored quickly, allowing business operations to continue with minimal downtime.

Conclusion

Cloud storage provides secure, durable, and cost-effective backup solutions while ensuring quick recovery during disasters.

3. Global Deployment Using CDN

Content Delivery Network (CDN)

A CDN stores copies of application content at multiple locations worldwide, allowing users to access data from the nearest server.

Edge Servers

Edge servers are located close to users and deliver content quickly, reducing network latency and improving application performance.

Global Availability

Cloud providers maintain data centers across multiple geographic regions. This ensures users receive services from the closest available location, improving speed and reliability.

Real-Time Example

A platform like Netflix or an online gaming application uses CDNs to stream videos or game updates from nearby edge servers, reducing buffering and lag for users worldwide.

Conclusion

CDNs and edge servers improve performance, reduce latency, and provide fast content delivery to global users.

4. Regulatory Compliance and Data Security in Cloud

Regulatory Compliance

Cloud providers support standards such as:

- GDPR
- ISO 27001
- PCI DSS
- HIPAA

These certifications help organizations meet legal and industry security requirements.

Logging and Monitoring

- Record all user activities and system events.
- Detect unauthorized access.
- Support security audits and compliance reporting.

Encryption

- Encrypt data at rest using AES-256.

- Encrypt data in transit using TLS/SSL.
- Protect encryption keys using key management services.

Data Sovereignty

Organizations can choose specific cloud regions to store data, ensuring information remains within the required country's legal jurisdiction.

Financial Example

A bank stores customer transaction data in an approved regional data center, encrypts all financial records, logs every transaction, and monitors suspicious activities to meet regulatory requirements.

Conclusion

Compliance, encryption, logging, and monitoring help financial institutions protect sensitive data while satisfying legal and regulatory standards.

5. Containerization and Kubernetes

Containers vs Virtual Machines

Feature	Containers	Virtual Machines
Startup Time	Seconds	Minutes
Resource Usage	Low	High
Performance	Faster	Slower
Operating System	Share host OS	Separate OS for each VM

Kubernetes Orchestration

Kubernetes automates:

- Container deployment
- Load balancing
- Auto-scaling
- Self-healing
- Rolling updates

This simplifies application management and ensures high availability.

Portability

Containers package the application with all its dependencies, allowing the same application to run consistently on different cloud providers without modification.

Microservices Example

An online shopping application can be divided into:

- User Service
- Product Service
- Order Service

- Payment Service
- Notification Service

Each service runs in its own container and is managed independently by Kubernetes.

Conclusion

Containers provide lightweight, fast, and portable deployments, while Kubernetes automates deployment, scaling, and management of modern cloud applications